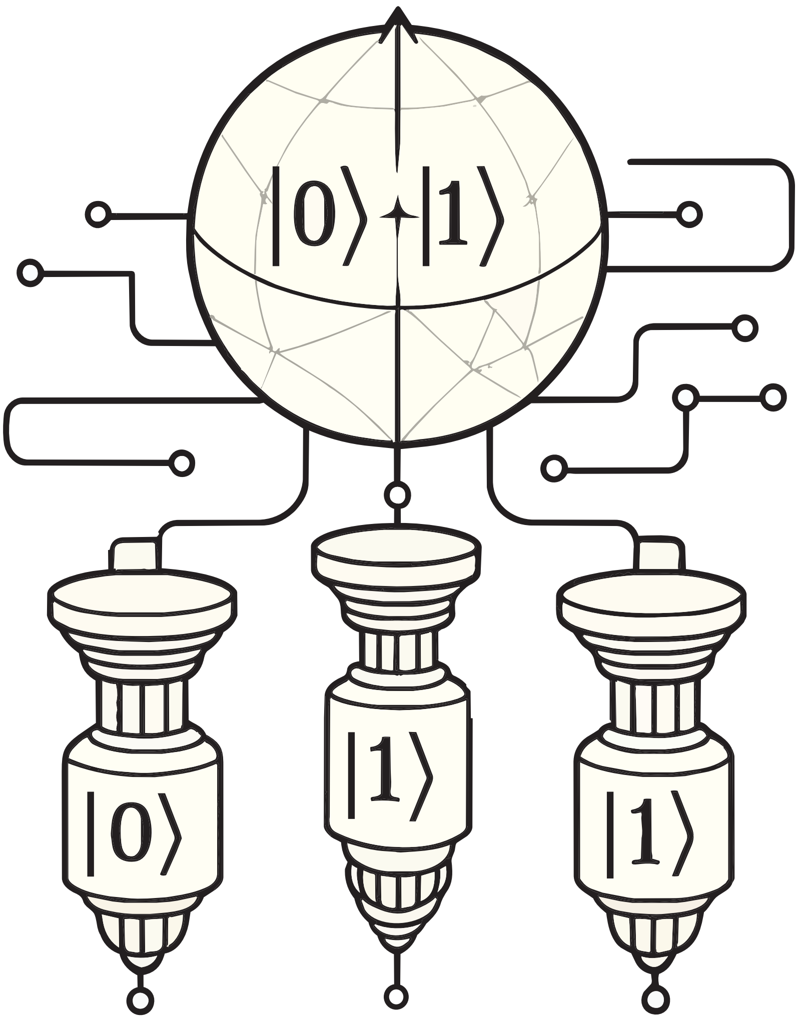
Quantum Computing: Beyond One Qubit

The central theme of this fourth article in the series on quantum computing is the Controlled-NOT (CNOT) gate, which is a multi-qubit quantum gate.



In the first three articles in this series on quantum computing, we developed an intuitive understanding of the behaviour of a single qubit using tools such as the Bloch sphere simulator. These visualisations helped us explore the action of various quantum gates that operate on a single qubit. We now move beyond single-qubit systems and begin our study of multi-qubit systems and the quantum gates that act on them.

However, before we proceed any further, let us take a brief detour to understand another formidable challenge in quantum computing. Developing an elegant quantum algorithm is only half the battle; the underlying quantum hardware must faithfully execute it. Unfortunately, this hardware exists in an extremely delicate quantum state, making it one of the greatest engineering challenges of our time. So far, our discussions have largely revolved around writing quantum programs and executing them on ideal simulators. Real quantum computers, however, are far from perfect and are susceptible to errors. Throughout this discussion, we shall continue focusing on superconducting quantum computers. In the previous article, we introduced the Josephson junction, the fundamental building block of a superconducting qubit. Let us now build upon that discussion and explore the enormous engineering challenge involved in constructing a superconducting quantum computer. Only then can we truly appreciate why today's quantum hardware is inherently prone to errors—and why even a brilliantly designed quantum algorithm can fail to deliver the expected results on real hardware.

Unlike a classical bit, which can reliably exist as either 0 or 1 under ordinary operating conditions, a superconducting qubit must preserve an extremely fragile quantum state. Even the slightest interaction with its surroundings—thermal radiation, stray electromagnetic fields, microscopic material defects, cosmic rays, or vibrations from the environment—can disturb this state, causing the qubit to lose the quantum information it is carrying. This process, known as decoherence, is one of the greatest